Exploratory Data Analysis

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```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(readr)
library(tidyr)
library(ggplot2)
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(here)
```

```
here() starts at /Users/owner/Desktop/Statistical Programming/armed_conflict/armed_conflict_TAS
```

The first thing I would explore is the distribution and types of variables. I can begin by simply looking at the data with head() or glimpse().

```
final_data <- read.csv(
  here("data", "processed", "final_data.csv"),
)
head(final_data)
```

```
  iso year maternal_mortality infant_mortality neonatal_mortality
1 AFG 2000               1450             90.5               60.9
2 AFG 2001               1390             87.9               59.7
3 AFG 2002               1300             85.3               58.5
4 AFG 2003               1240             82.7               57.2
5 AFG 2004               1180             80.0               55.9
6 AFG 2005               1140             77.3               54.6
  under5_mortality drought earthquake best armed_conflict  gdp_1000 oecd
1            129.2       1          0 5065              1        NA    0
2            125.2       0          1 5394              1        NA    0
3            121.1       0          1 5553              1 0.1835328    0
4            116.9       0          1 1157              1 0.2004626    0
5            112.6       0          1  944              1 0.2216576    0
6            108.4       0          1  817              1 0.2550551    0
  pop_dens    urban  age_dep male_edu     temp rainfall_1000
1 14.13654 16.25324 108.3466 2.762086 12.69959     0.2763704
2 14.23156 16.25661 108.9899 2.856936 12.85570     0.2793079
3 14.32270 16.42654 109.3472 2.954241 12.71081     0.3805710
4 14.40691 16.60701 109.4475 3.054121 12.16592     0.4288939
5 15.21947 16.71367 109.2868 3.156706 13.04643     0.3754336
6 15.33619 16.85096 107.9646 3.262133 12.23141     0.4415680
```

```
glimpse(final_data)
```

```
Rows: 3,732
Columns: 18
$ iso                <chr> "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "A~
$ year               <int> 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 200~
$ maternal_mortality <int> 1450, 1390, 1300, 1240, 1180, 1140, 1120, 1090, 103~
$ infant_mortality   <dbl> 90.5, 87.9, 85.3, 82.7, 80.0, 77.3, 74.6, 71.9, 69.~
$ neonatal_mortality <dbl> 60.9, 59.7, 58.5, 57.2, 55.9, 54.6, 53.2, 51.7, 50.~
$ under5_mortality   <dbl> 129.2, 125.2, 121.1, 116.9, 112.6, 108.4, 104.1, 99~
$ drought            <int> 1, 0, 0, 0, 0, 0, 1, NA, 1, 0, 0, 1, 0, 0, NA, 0, N~
$ earthquake         <int> 0, 1, 1, 1, 1, 1, 1, NA, 0, 1, 1, 0, 1, 1, NA, 1, N~
```

```
$ best               <int> 5065, 5394, 5553, 1157, 944, 817, 1711, 4982, 7020,~
$ armed_conflict     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ gdp_1000           <dbl> NA, NA, 0.1835328, 0.2004626, 0.2216576, 0.2550551,~
$ oecd               <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ pop_dens           <dbl> 14.136539, 14.231558, 14.322705, 14.406906, 15.2194~
$ urban              <dbl> 16.25324, 16.25661, 16.42654, 16.60701, 16.71367, 1~
$ age_dep            <dbl> 108.34663, 108.98989, 109.34716, 109.44753, 109.286~
$ male_edu           <dbl> 2.762086, 2.856936, 2.954241, 3.054121, 3.156706, 3~
$ temp               <dbl> 12.69959, 12.85570, 12.71081, 12.16592, 13.04643, 1~
$ rainfall_1000      <dbl> 0.2763704, 0.2793079, 0.3805710, 0.4288939, 0.37543~
```

Then I would generate summary statistics for numeric variables (such as maternal mortality ratio, newborn mortality, etc.) using the summary() function.

```
final_data |>
    summary()
```

```
     iso                 year      maternal_mortality infant_mortality
 Length:3732        Min.   :2000   Min.   :   2.0     Min.   :  1.60
 Class :character   1st Qu.:2005   1st Qu.:  17.0     1st Qu.:  7.60
 Mode  :character   Median :2010   Median :  66.0     Median : 18.90
                    Mean   :2009   Mean   : 210.6     Mean   : 28.90
                    3rd Qu.:2014   3rd Qu.: 299.8     3rd Qu.: 44.52
                    Max.   :2019   Max.   :2480.0     Max.   :138.10
                                   NA's   :438        NA's   :32
 neonatal_mortality under5_mortality    drought        earthquake
 Min.   : 0.80      Min.   :  2.00   Min.   :0.000   Min.   :0.0000
 1st Qu.: 4.90      1st Qu.:  9.00   1st Qu.:0.000   1st Qu.:0.0000
 Median :12.10      Median : 22.20   Median :1.000   Median :1.0000
 Mean   :16.18      Mean   : 40.50   Mean   :0.545   Mean   :0.5333
 3rd Qu.:25.32      3rd Qu.: 61.33   3rd Qu.:1.000   3rd Qu.:1.0000
 Max.   :60.90      Max.   :224.90   Max.   :1.000   Max.   :1.0000
 NA's   :32         NA's   :32       NA's   :3132    NA's   :3132
      best         armed_conflict      gdp_1000            oecd
 Min.   :    0.0   Min.   :0.0000   Min.   :  0.1105   Min.   :0.000
 1st Qu.:   13.0   1st Qu.:0.0000   1st Qu.:  1.2383   1st Qu.:0.000
 Median :   99.0   Median :1.0000   Median :  4.0719   Median :0.000
 Mean   : 1311.9   Mean   :0.6875   Mean   : 11.4917   Mean   :0.171
 3rd Qu.:  761.2   3rd Qu.:1.0000   3rd Qu.: 13.1531   3rd Qu.:0.000
 Max.   :78644.0   Max.   :1.0000   Max.   :123.6787   Max.   :1.000
```

```
 NA's   :2708      NA's   :2708     NA's   :74         NA's   :12
   pop_dens          urban            age_dep          male_edu
 Min.   : 0.00   Min.   : 0.1025   Min.   : 16.17   Min.   : 1.067
 1st Qu.:14.79   1st Qu.:17.2872   1st Qu.: 47.94   1st Qu.: 5.904
 Median :27.52   Median :30.2535   Median : 55.51   Median : 8.368
 Mean   :30.57   Mean   :30.6948   Mean   : 61.94   Mean   : 8.258
 3rd Qu.:40.72   3rd Qu.:41.6558   3rd Qu.: 77.11   3rd Qu.:10.849
 Max.   :99.86   Max.   :93.4135   Max.   :111.48   Max.   :14.441
 NA's   :32      NA's   :32        NA's   :12       NA's   :32
      temp         rainfall_1000
 Min.   :-2.405   Min.   :0.01993
 1st Qu.:12.928   1st Qu.:0.59146
 Median :21.958   Median :1.01288
 Mean   :19.625   Mean   :1.20216
 3rd Qu.:25.869   3rd Qu.:1.68706
 Max.   :29.676   Max.   :4.71081
 NA's   :32       NA's   :32
```

To get a better understanding of the limits of these summary statistics, I could randomly remove values of the statistic for certain country-years and then see how the summary statistics change depending on the group of country-years with missing data.

```r
set.seed(123)

# Choose the statistic to test and the percentage removed in each simulation.
outcome <- "maternal_mortality"
removal_proportion <- 0.20

# Restrict selection to country-years with an observed outcome value.
available_country_years <- final_data |>
  filter(!is.na(.data[[outcome]])) |>
  distinct(iso, year)

n_to_remove <- floor(nrow(available_country_years) * removal_proportion)

# Draw two different, non-overlapping sets of country-year observations.
removed_set_1 <- available_country_years |>
  slice_sample(n = n_to_remove) |>
  mutate(removal_set = "Set 1")
```

```r
removed_set_2 <- available_country_years |>
  anti_join(removed_set_1, by = c("iso", "year")) |>
  slice_sample(n = n_to_remove) |>
  mutate(removal_set = "Set 2")

removed_country_years <- bind_rows(removed_set_1, removed_set_2) |>
  arrange(removal_set, iso, year)

# Display exactly which country-years are removed in each simulation.
removed_country_years |>
  group_by(removal_set, iso)
```

```
# A tibble: 1,316 x 3
# Groups:   removal_set, iso [358]
   iso    year removal_set
   <chr> <int> <chr>
 1 AFG    2007 Set 1
 2 AFG    2009 Set 1
 3 AFG    2012 Set 1
 4 AFG    2015 Set 1
 5 AGO    2005 Set 1
 6 AGO    2012 Set 1
 7 ALB    2004 Set 1
 8 ALB    2012 Set 1
 9 ALB    2015 Set 1
10 ARE    2016 Set 1
# i 1,306 more rows
```

```r
remove_country_years <- function(data, removal_set, outcome) {
  data |>
    left_join(
      removal_set |>
        select(iso, year) |>
        mutate(remove_value = TRUE),
      by = c("iso", "year")
    ) |>
    mutate(
      simulated_value = if_else(
        coalesce(remove_value, FALSE),
        NA_real_,
        .data[[outcome]]
      )
    ) |>
```

```
    select(-remove_value)
}

summarise_values <- function(data, value, scenario, n_removed) {
  data |>
    summarise(
      scenario = scenario,
      n_removed = n_removed,
      n_observed = sum(!is.na(.data[[value]])),
      mean = mean(.data[[value]], na.rm = TRUE),
      median = median(.data[[value]], na.rm = TRUE),
      sd = sd(.data[[value]], na.rm = TRUE)
    )
}

simulated_set_1 <- remove_country_years(final_data, removed_set_1, outcome)
simulated_set_2 <- remove_country_years(final_data, removed_set_2, outcome)

comparison <- bind_rows(
  summarise_values(final_data, outcome, "Original data", 0),
  summarise_values(simulated_set_1, "simulated_value", "Set 1 removed", n_to_remove),
  summarise_values(simulated_set_2, "simulated_value", "Set 2 removed", n_to_remove)
)

comparison
```

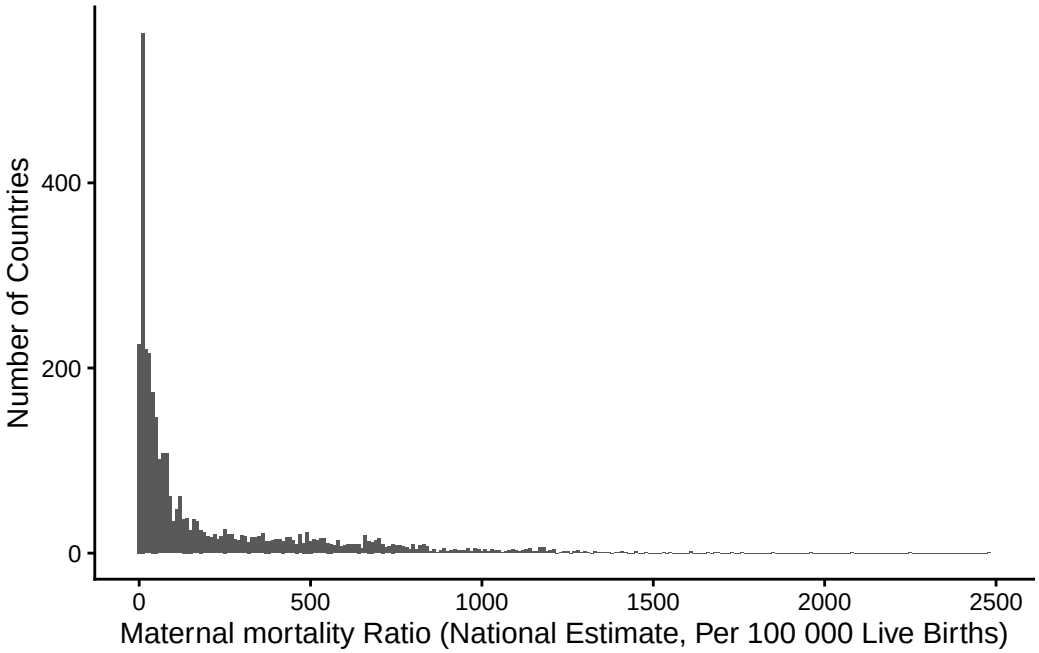
```
       scenario n_removed n_observed     mean median       sd
1 Original data         0       3294 210.6178   66.0 303.7754
2 Set 1 removed       658       2636 209.5508   65.0 303.2183
3 Set 2 removed       658       2636 206.3437   64.5 299.5825
```

I could also use graphs to map the distribution of variables with a logarithmic transformation of the y-values where appropriate.

```
#distribution of maternal mortality
final_data |>
  ggplot(aes(x = maternal_mortality)) +
  geom_histogram(binwidth = 10) +
  theme_classic() +
  labs(
    x = "Maternal mortality Ratio (National Estimate, Per 100 000 Live Births)",
```

```
    y = "Number of Countries"
  )
```

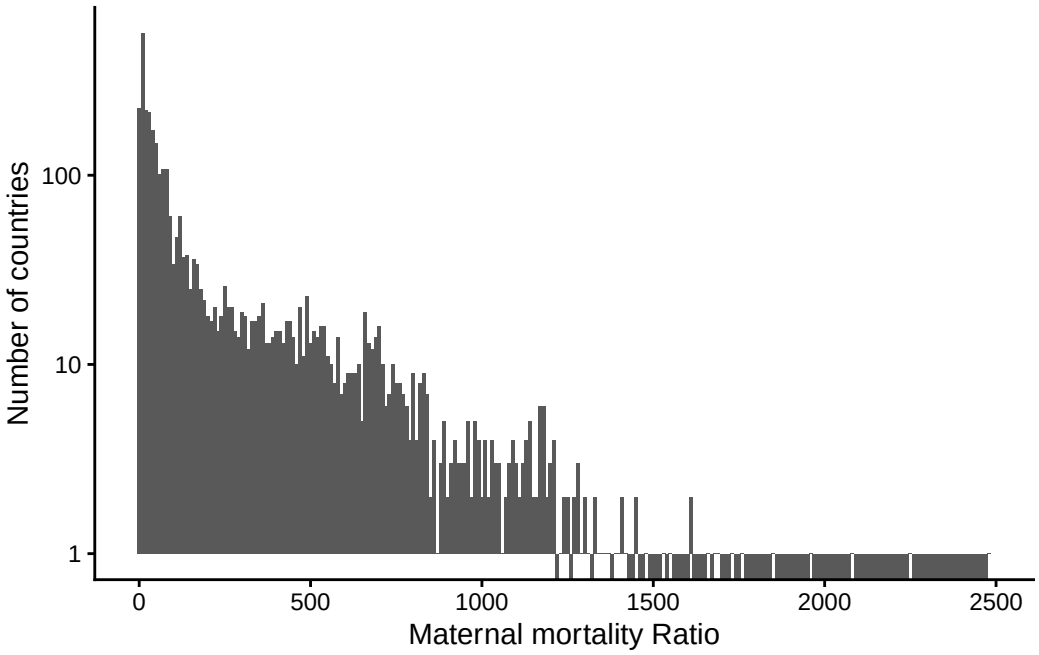
```
Warning: Removed 438 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
final_data |>#log tranformation
ggplot(aes(x = maternal_mortality)) +
geom_histogram(binwidth = 10) +
theme_classic() +
labs(
  x = "Maternal mortality Ratio",
  y = "Number of countries"
) +
scale_y_log10()
```

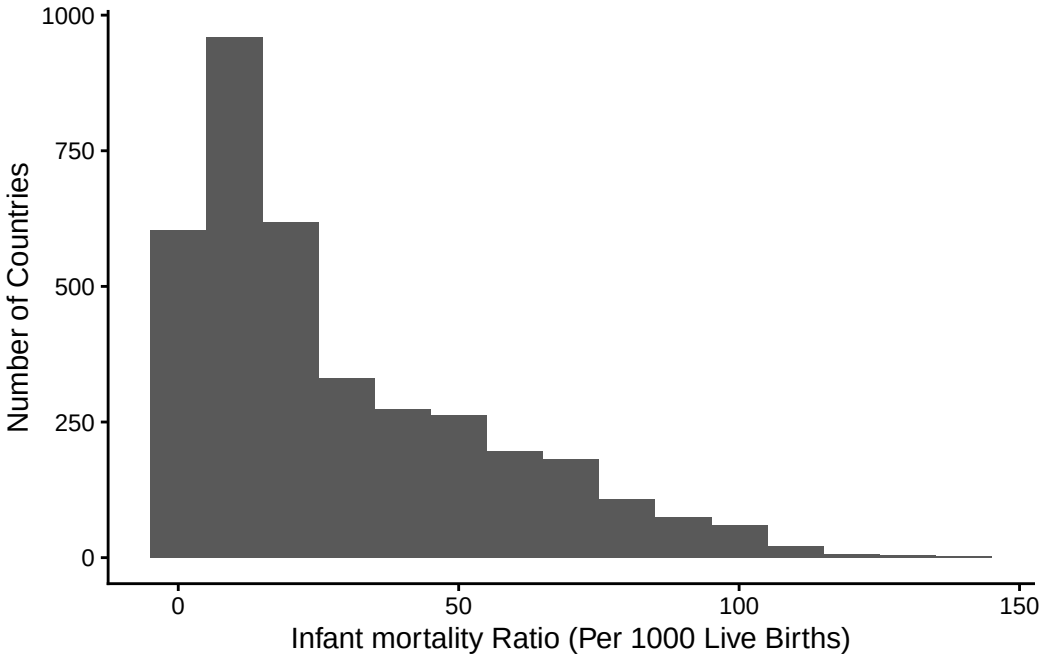
```
Warning: Removed 438 rows containing non-finite outside the scale range
(`stat_bin()`).
```

```
Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```



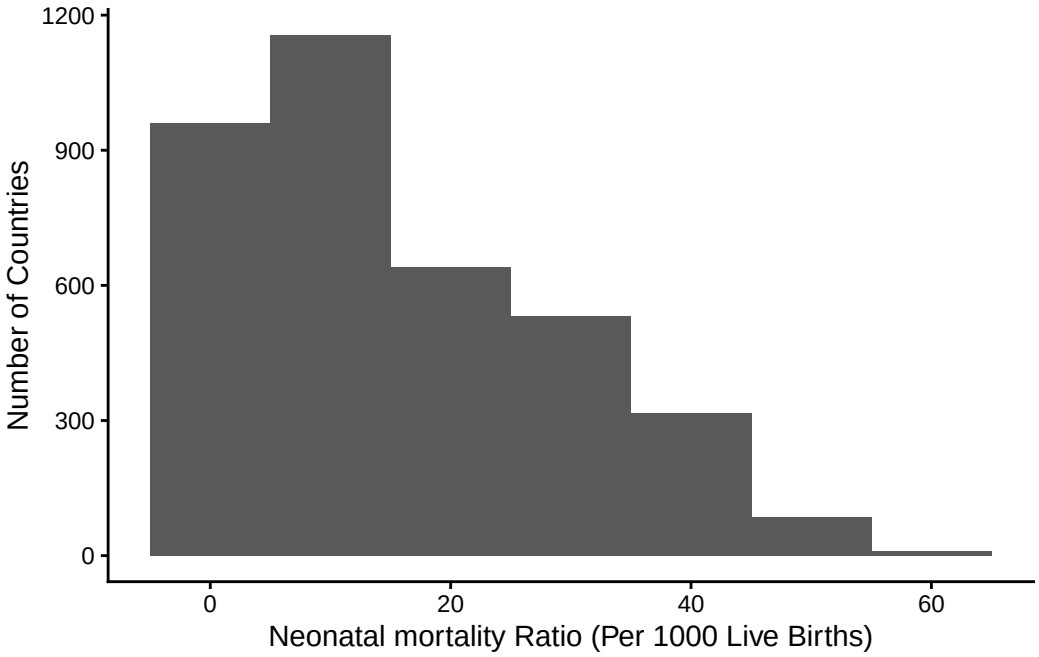

```
#distribution of infant mortality ratio
final_data |>
  ggplot(aes(x = infant_mortality)) +
  geom_histogram(binwidth = 10) +
  theme_classic() +
  labs(
    x = "Infant mortality Ratio (Per 1000 Live Births)",
    y = "Number of Countries"
  )
```

```
Warning: Removed 32 rows containing non-finite outside the scale range
(`stat_bin()`).
```



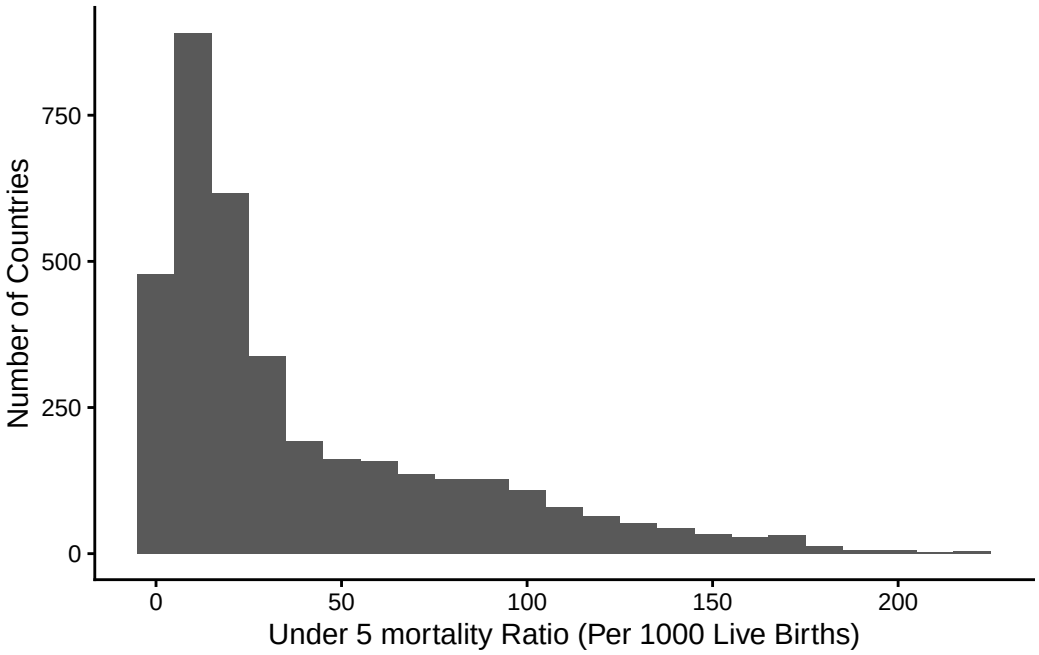

```
#distribution of neonatal mortality ratio
final_data |>
  ggplot(aes(x = neonatal_mortality)) +
  geom_histogram(binwidth = 10) +
  theme_classic() +
  labs(
    x = "Neonatal mortality Ratio (Per 1000 Live Births)",
    y = "Number of Countries"
  )
```

```
Warning: Removed 32 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
#distribution of under 5 mortality ratio
final_data |>
  ggplot(aes(x = under5_mortality)) +
  geom_histogram(binwidth = 10) +
  theme_classic() +
  labs(
    x = "Under 5 mortality Ratio (Per 1000 Live Births)",
    y = "Number of Countries"
  )
```

```
Warning: Removed 32 rows containing non-finite outside the scale range
(`stat_bin()`).
```

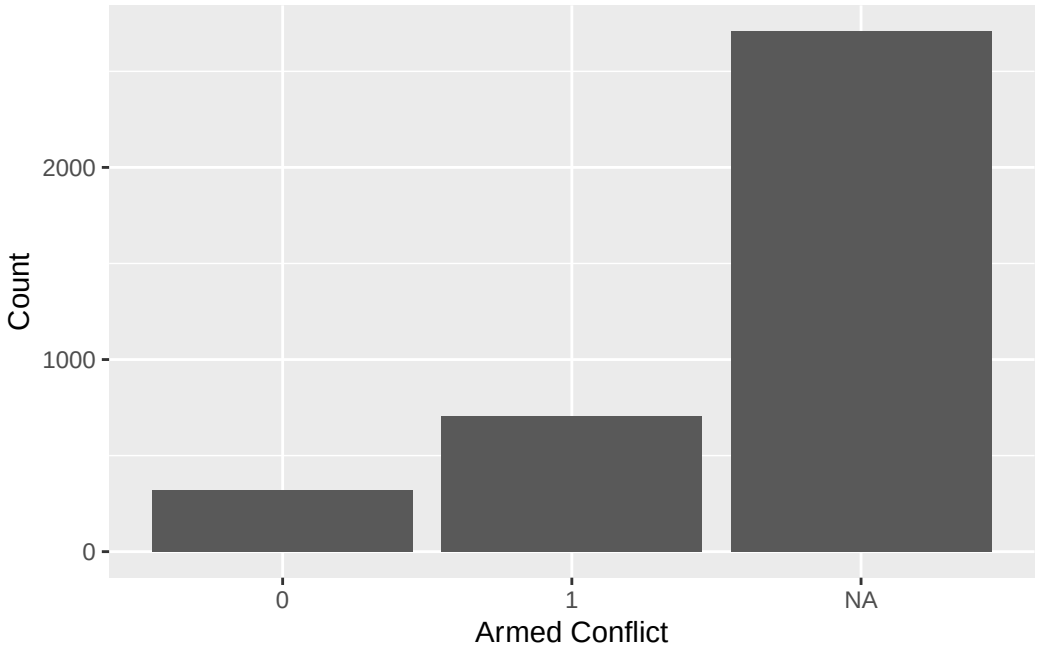



```
#There are a couple of outliers with maternal mortality values> 1750 per 100 000 live births
final_data|>
    filter(maternal_mortality>1750)|>
      select(iso, maternal_mortality, year)
```

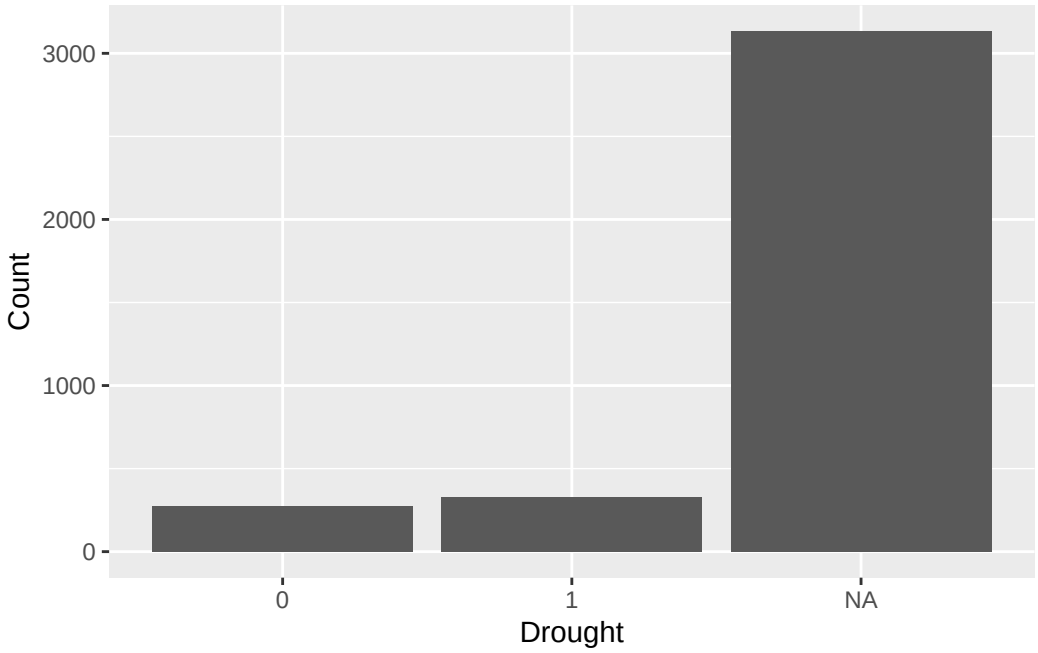
```
  iso maternal_mortality year
1 SLE               2480 2000
2 SLE               2250 2001
3 SLE               2080 2002
4 SLE               1960 2003
5 SLE               1850 2004
6 SLE               1760 2005
```

```
#SLE (Sierra Lione) has a high maternal mortality rate from 2000-2005 there likely was a long-las

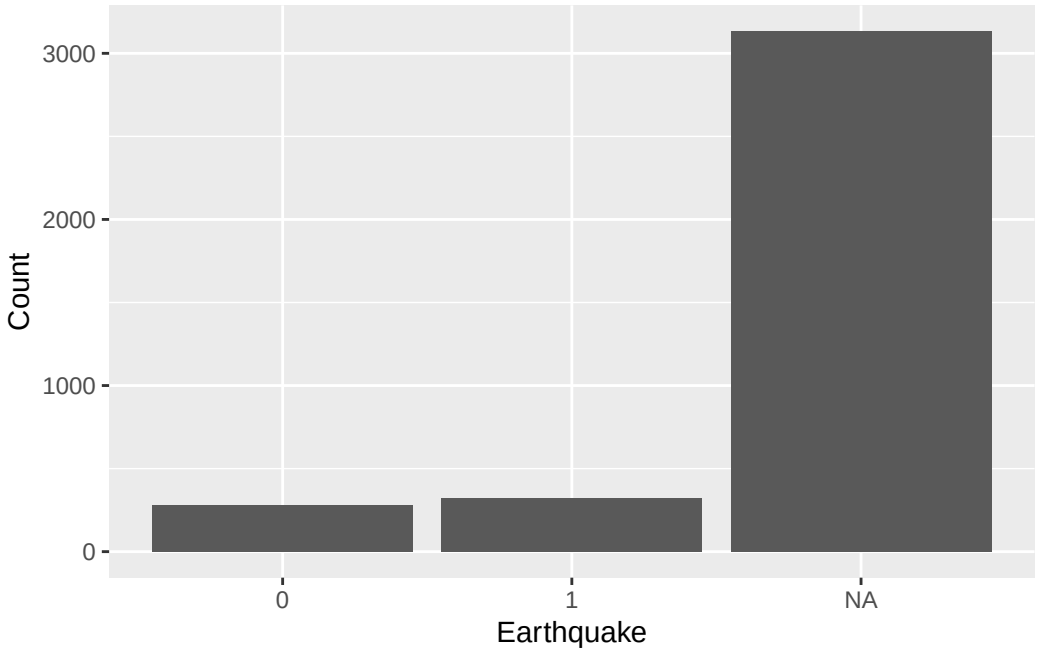
#distribution of armed conflict
final_data|>
  ggplot() +
    geom_bar(aes(
        x=factor(armed_conflict)
    )) +
    labs(x = "Armed Conflict", y = "Count")
```




```
#drought
  final_data|>
  ggplot() +
    geom_bar(aes(
        x=factor(drought)
    )) +
    labs(x = "Drought", y = "Count")
```



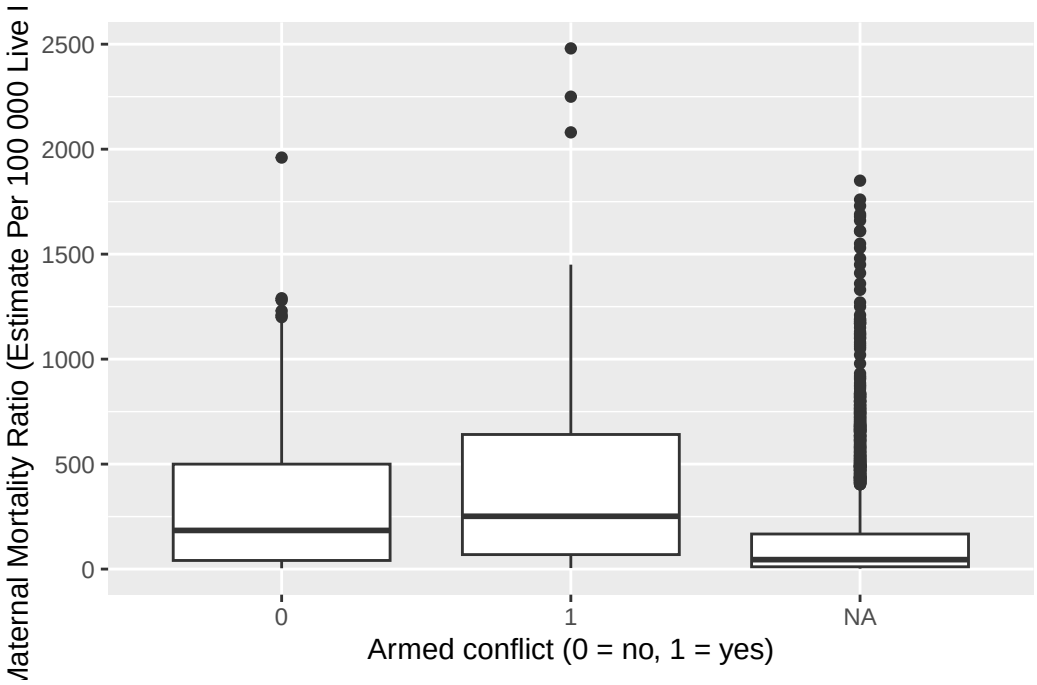

```
#earthquake
  final_data|>
  ggplot() +
    geom_bar(aes(
        x=factor(earthquake)
    )) +
    labs(x = "Earthquake", y = "Count")
```



This may help identify outliers, which I could explore further for patterns with other variables.

```
#Looking at armed conflict and maternal mortality together
 ggplot(final_data, aes(x = factor(armed_conflict), y = maternal_mortality)) +
  geom_boxplot() +
  labs(
    x = "Armed conflict (0 = no, 1 = yes)",
    y = "Maternal Mortality Ratio (Estimate Per 100 000 Live Births)"
  )
```

```
Warning: Removed 438 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




```
#The presence of armed conflict seems associated with the highest recorded maternal maortality ra
```

I could look at the variable types using the table() function and see where expected variable types do not match.

```
#Note: opted not to use the table function, instead directly used class()
final_data |>
  summarise(
    across(everything(), ~ paste(class(.x), collapse = ", "))
  ) |>
  pivot_longer(
    cols = everything(),
    names_to = "variable",
    values_to = "class"
  )
```

```
# A tibble: 18 x 2
   variable           class
   <chr>              <chr>
 1 iso                character
 2 year               integer
```

```
 3 maternal_mortality integer
 4 infant_mortality   numeric
 5 neonatal_mortality numeric
 6 under5_mortality   numeric
 7 drought            integer
 8 earthquake         integer
 9 best               integer
10 armed_conflict     integer
11 gdp_1000           numeric
12 oecd               integer
13 pop_dens           numeric
14 urban              numeric
15 age_dep            numeric
16 male_edu           numeric
17 temp               numeric
18 rainfall_1000      numeric
```

Once I am done observing the kinds and distributions of the variables, I would explore the degree, extent, and kinds of missingness, as the preliminary glimpse into the data would reveal that there is quite a bit of missingness in the armed conflict data.

```
get_dupes(final_data)#checking for duplicate data
```

```
No variable names specified - using all columns.

No duplicate combinations found of: iso, year, maternal_mortality, infant_mortality, neonatal_mor

 [1] iso               year              maternal_mortality infant_mortality
 [5] neonatal_mortality under5_mortality drought            earthquake
 [9] best              armed_conflict    gdp_1000           oecd
[13] pop_dens          urban             age_dep            male_edu
[17] temp              rainfall_1000     dupe_count
<0 rows> (or 0-length row.names)
```

```
final_data |>#look at degree of missingness for each variable
  summarise(
    across(everything(), ~ sum(is.na(.x)))
  ) |>
  pivot_longer(
```

```
    cols = everything(),
    names_to = "variable",
    values_to = "n_missing"
  ) |>
  mutate(percent_missing = 100 * n_missing / nrow(final_data)) |>
  arrange(desc(n_missing))
```

```
# A tibble: 18 x 3
   variable           n_missing percent_missing
   <chr>                  <int>           <dbl>
 1 drought                 3132          83.9
 2 earthquake              3132          83.9
 3 best                    2708          72.6
 4 armed_conflict          2708          72.6
 5 maternal_mortality       438          11.7
 6 gdp_1000                  74           1.98
 7 infant_mortality          32           0.857
 8 neonatal_mortality        32           0.857
 9 under5_mortality          32           0.857
10 pop_dens                  32           0.857
11 urban                     32           0.857
12 male_edu                  32           0.857
13 temp                      32           0.857
14 rainfall_1000             32           0.857
15 oecd                      12           0.322
16 age_dep                   12           0.322
17 iso                        0           0
18 year                       0           0
```

```
#the same amount of missing values for drought and earthquake- worth investigating
final_data|>
  tabyl(drought, earthquake,  show_na = TRUE)
```

```
 drought   0   1  NA_
       0   0 273    0
       1 280  47    0
      NA   0   0 3132
```

```
#earthquake and drought seem to be missing together
```